LittleStory iOS HLD

LittleStory iOS — High-level design

Status: Draft v0.2 for owner review · 11 Nov 2026 (v0.2: iPhone + iPad universal app, exact latest versions, minimum iOS 26) **Scope:** the native iOS app for LittleStory, in a new repository (proposed name `LittleStory-iOS`). **Companion:** Low-level design (classes, files, data, sequences). **Reference:** the Android app (`shivayogih/LittleStory`, `development` @ `5c505c6`) and its HLD/LLD. The iOS app copies the product, the screens, the content and the rules, not the Kotlin code.

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0. Summary in eleven points

1. **Native Swift 6.4 + SwiftUI** app, built with **Xcode 27.2** against the **iOS 27.2 SDK**, minimum **iOS / iPadOS 26**. No cross-platform framework (decided 11 Nov 2026, §3).
2. **Clean, layered architecture**: Presentation → Domain → Data, with the Domain layer as pure Swift that knows nothing about UI, storage or the network.
3. **SOLID** throughout: every dependency is a Swift protocol, one composition root wires real or mock implementations, and each type has one job.
4. **Modular**: one local Swift package (`LittleStoryKit`) with a target per layer and per feature, mirroring the Android modules one for one. A feature can never import another feature or the Data layer.
5. **Same product as Android**: 21 screens, five tabs, the same flows, the same mock stories in **English and Kannada**, the same parental gate.
6. **Zero paid services**: no backend, no AI API, no analytics, no ads, no paid SDKs. Every dependency is first-party Apple or free open source.
7. **AI is mocked**: story writing, narration, pictures, downloads and subscription are deterministic simulations behind protocols, labelled in the UI, and swappable later without touching screens.
8. **Languages are data**: English is default and fallback, Kannada is complete, and a new language is one registry entry plus translation files.
9. **Quality gates from day one**: Swift Testing unit tests, snapshot tests in EN / KN / largest Dynamic Type, UI tests, SwiftLint, SwiftFormat, a hardcoding check and a translation check, all run by GitHub Actions on every pull request.

10. **One universal app for iPhone and iPad**: every screen adapts to compact and regular widths, portrait and landscape, iPad split screen and freely resized windows (§6A).
11. **Child safety first**: all data stays on the device, no tracking, a parental gate before parent-only screens, and a privacy manifest, so the app can later qualify for the App Store Kids category.

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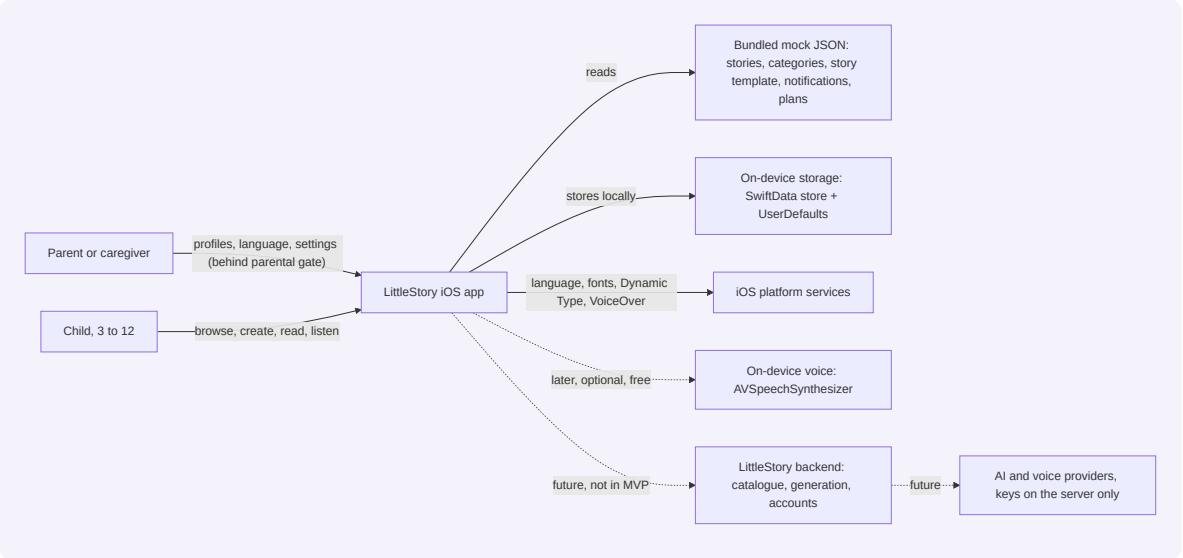
1. What the app is

LittleStory is a storytelling app for children aged 3–12 and their parents. A family browses a bundled catalogue of stories, creates a personalised story from a few words about the child's day, reads it page by page or listens to it in a narration player, and manages child profiles, language and parent settings.

In the iOS MVP	Not in the iOS MVP
Swift 6, SwiftUI, Observation, Swift Concurrency	Real AI provider, backend, sign-in, cloud sync
21 screens, same as Android (§6)	Real payments (StoreKit purchases), real push notifications
Mock content in English and Kannada (same JSON as Android)	Analytics, crash-reporting SaaS, advertising
Simulated story generation, narration, downloads, subscription	Any network call; the app has no server to call
Local storage: SwiftData + UserDefaults	Accounts, login/logout (follows the Android plan in PR #20)
Parental gate before parent-only screens	Widgets, Apple Watch, Mac and Vision Pro builds
Universal app: iPhone and iPad, portrait and landscape, iPad split screen and resizable windows (§6A)	Store submission (owner only, and it needs a paid account, §12)

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2. System context



- Solid lines exist in the MVP. Dotted lines are replacement points only.
- The app holds no API keys and makes no network requests (the same rule as Android NFR-07).

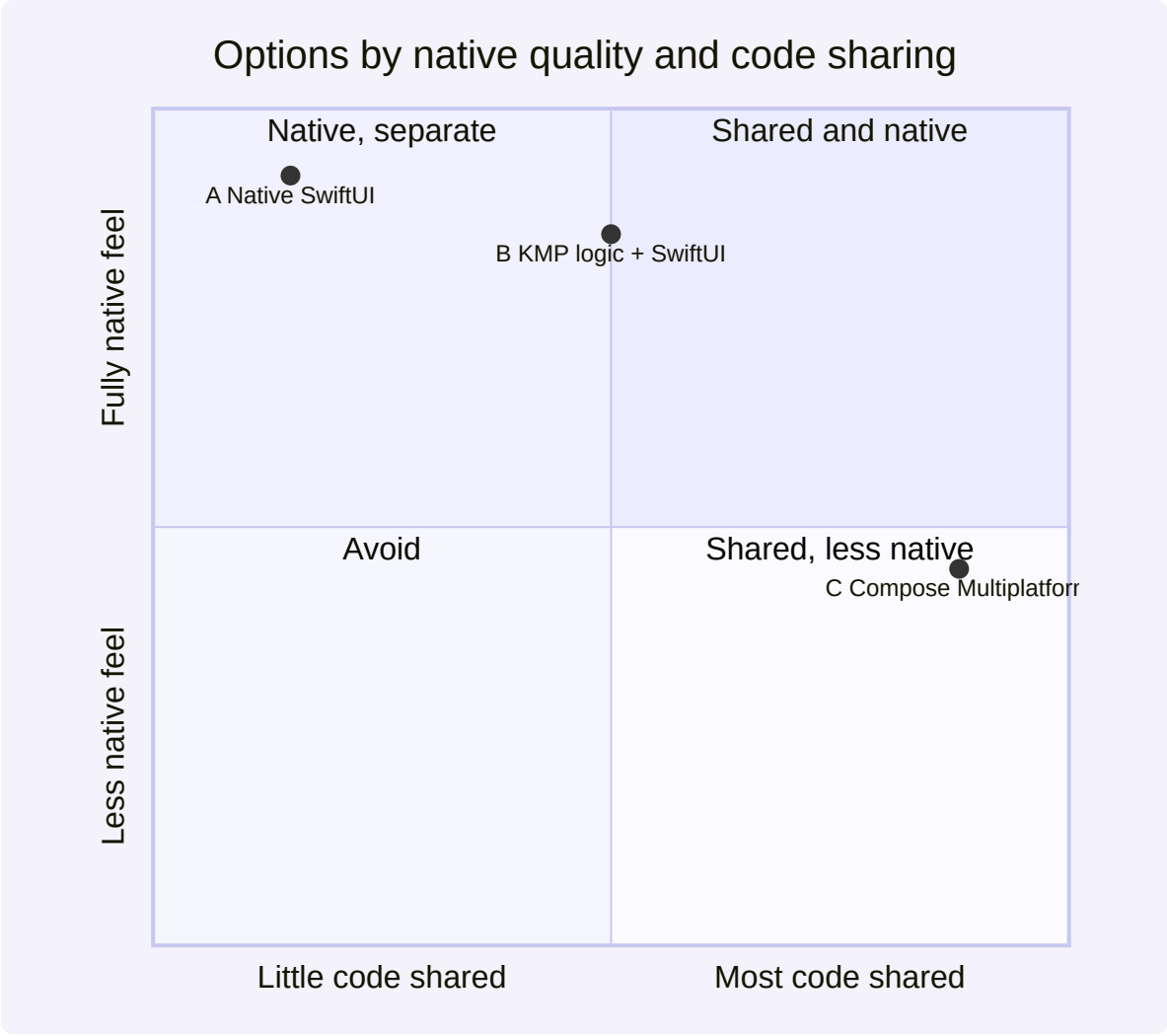
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3. Key decision: native SwiftUI or Kotlin/Compose Multiplatform?

3.1 Options

	A. Native Swift + SwiftUI (recommended)	**B. Kotlin Multiplatform (shared logic) + SwiftUI**	**C. Compose Multiplatform (shared logic and UI)**
What is shared with Android	Contracts only: mock JSON, design tokens, string keys, test cases	`:domain`, `:core:common`, part of `:data`	Almost everything
UI quality on iOS	Fully native: navigation, gestures, VoiceOver, Dynamic Type	Fully native	Compose look on iOS; accessibility and text input still maturing
Fits "a new repo for iOS"	Yes, fully independent	No: the shared code must live in the Android repo or a third repo, so the two builds are coupled	No: it is really a change to the Android repo
Android-only pieces to rewrite	All, but the domain is small (≈ 15 models, 11 repository interfaces, 2 simulations)	Hilt, `R.string` / `UiText`, `AppCompatDelegate` locales, asset loading, Room/DataStore wiring	Same as B, plus UI interop work

	A. Native Swift + SwiftUI (recommended)	B. Kotlin Multiplatform (shared logic) + SwiftUI	C. Compose Multiplatform (shared logic and UI)
Hiring and community	Largest iOS talent pool and tooling	Needs both Kotlin and Swift skills	Smallest
Build and CI	Xcode only	Gradle + Xcode, framework export, slower CI	Gradle + Xcode
Risk	Two codebases can drift	Toolchain coupling, Swift interop friction	Platform feel, accessibility



3.2 Recommendation: Option A, native Swift + SwiftUI

1. The owner asked for a separate iOS repository; only Option A keeps it truly independent.
2. The Android audit shows the parts that are hard to share are exactly the ones that matter most for this app: localization (`R.string`, `UiText`, `AppCompatDelegate`), dependency injection (Hilt) and asset loading. With KMP those would be rewritten anyway.
3. The shared Kotlin logic is small and stable. Rewriting it in Swift is a few days of work, and the Android unit tests become the iOS acceptance tests.

4. Children's apps live or die on accessibility, text in Kannada and a native feel. SwiftUI gives first-class VoiceOver, Dynamic Type and system fonts.
5. Drift is controlled by **sharing contracts, not code** (§10): the same mock JSON files, the same design tokens, the same translation keys, the same test case IDs.
6. The door stays open: because Domain is a pure Swift package, moving to KMP later would replace one target, not the app.

> **Owner decision D-iOS-1 (11 Nov 2026): Option A, native Swift + SwiftUI.**
> Everything below follows it.

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4. Technology stack

All free. Versions below are the **newest stable releases on 11 Nov 2026** (checked on xcodereleases.com, Swift Package Index and GitHub release pages). They are pinned in the repository (`.xcode-version` , `Package.resolved` , `.github/workflows`), and Renovate-style update PRs (or a monthly manual check) keep them current; one major toolchain upgrade per PR.

4.1 Toolchain and platform

Item	Version	Notes
Xcode	**27.2** (27B52, 29 Oct 2026)	needs macOS 26.6 or newer on the build Mac
Swift	**6.4.2** (bundled with Xcode 27.2)	Swift 6 language mode, complete strict concurrency, warnings as errors in CI
iOS / iPadOS SDK	**27.2**	build against the newest SDK
Minimum deployment	**iOS 26.0 / iPadOS 26.0**	same devices as iOS 27 (iPhone 11 / SE 2nd gen and newer; iPad 8th gen, Air 3rd gen, mini 5th gen, Pro 11-inch 1st gen and newer); native Liquid Glass, current SwiftUI and SwiftData APIs without availability checks
Devices	iPhone and iPad, one universal target	§6A
CI runner	GitHub Actions `macos-26` (arm64)	its default Xcode is 26.6, so the workflow selects Xcode 27.2 explicitly (LLD §14.2)

4.2 Frameworks and libraries

Area	Choice	Version	Why
UI	SwiftUI (UIKit only where SwiftUI has no API)	SDK 27.2	Declarative, adaptive layouts, Liquid Glass by default
State	Observation (`@Observable`) view models on `@MainActor` , immutable `State` structs	SDK	Fine-grained updates, no Combine
Async	Swift Concurrency (`async/await` , `AsyncStream` , actors), injected `Clock`	Swift 6.4	Deterministic tests
Async helpers	`swift-async-algorithms` (Apple, Apache 2.0)	**1.1.7**	`combineLatest` , `debounce`
Navigation	`TabView` with `.sidebarAdaptable` , `NavigationSplitView` on regular width, `NavigationStack` + typed `Route` enums	SDK	Tab bar on iPhone, sidebar on iPad (§6A)
DI	Constructor injection + one composition root (`AppContainer`)	—	No framework needed
Persistence	**SwiftData** behind repository protocols	SDK	First-party, versioned schema
Persistence fallback	GRDB.swift (MIT), only if SwiftData blocks us (LLD §4.4)	**7.12.0**	Mature SQLite, mirrors Android Room
Preferences	`UserDefaults` wrapped in an actor	SDK	Same keys as Android DataStore
Content	Bundled JSON + `Codable`	SDK	Same files as Android
Localization	String Catalogs (`.xcstrings`), `LocalizedStringResource`	SDK	Plurals, completeness check
Fonts	Noto Sans Kannada (SIL OFL) + SF Pro	—	Kannada shaping at all sizes
Logging	`os.Logger`	SDK	Free, private by default
Unit tests	**Swift Testing**	bundled with Swift 6.4	Parallel, parameterised
Test clock	`swift-clocks` (Point-Free, MIT)	**1.1.1**	`TestClock` , `ImmediateClock`
Snapshot tests	`swift-snapshot-testing` (Point-Free, MIT)	**1.19.7**	iPhone + iPad × EN / KN images
UI tests	XCUITest	Xcode 27.2	Accessibility identifiers
Lint	SwiftLint (MIT)	**0.65.1**	Rule gate
Format	SwiftFormat (MIT)	**0.64.6**	Style gate

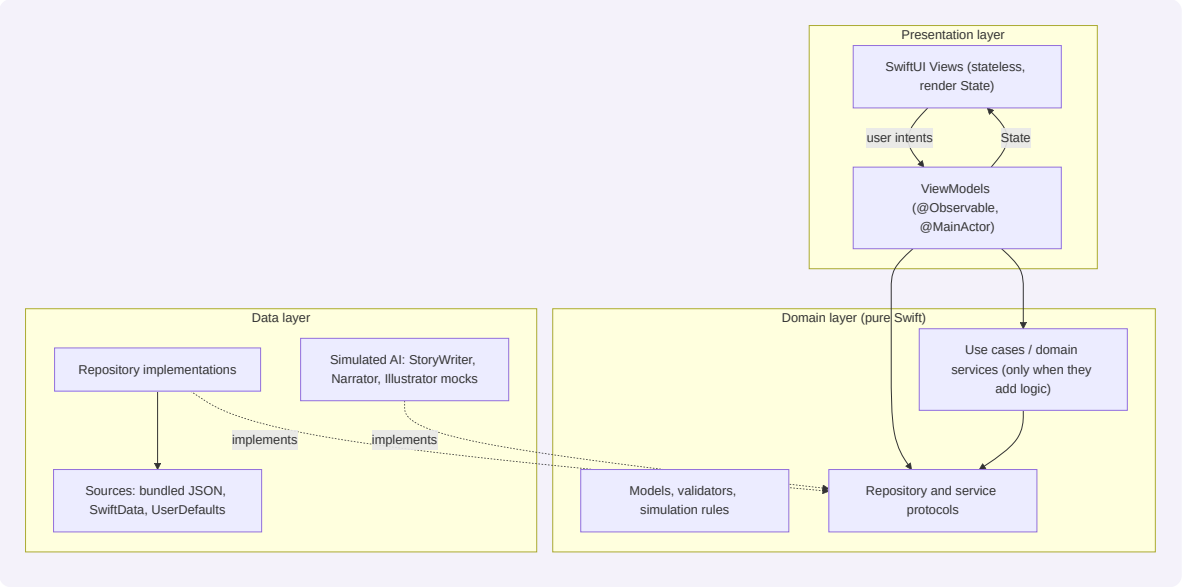
Area	Choice	Version	Why
CI actions	`actions/checkout`, `actions/upload-artifact`	**v7**	Newest majors
Project file	Thin `.xcodeproj`; all code in the local package	—	Avoids `project.pbxproj` conflicts

Deliberately not used: Firebase, Crashlytics, any analytics SDK, RevenueCat, The Composable Architecture (good, but a large learning curve for a small team; our pattern in §7 gives the same unidirectional flow with no dependency), Combine for new code, CocoaPods.

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5. Architecture

5.1 Layers

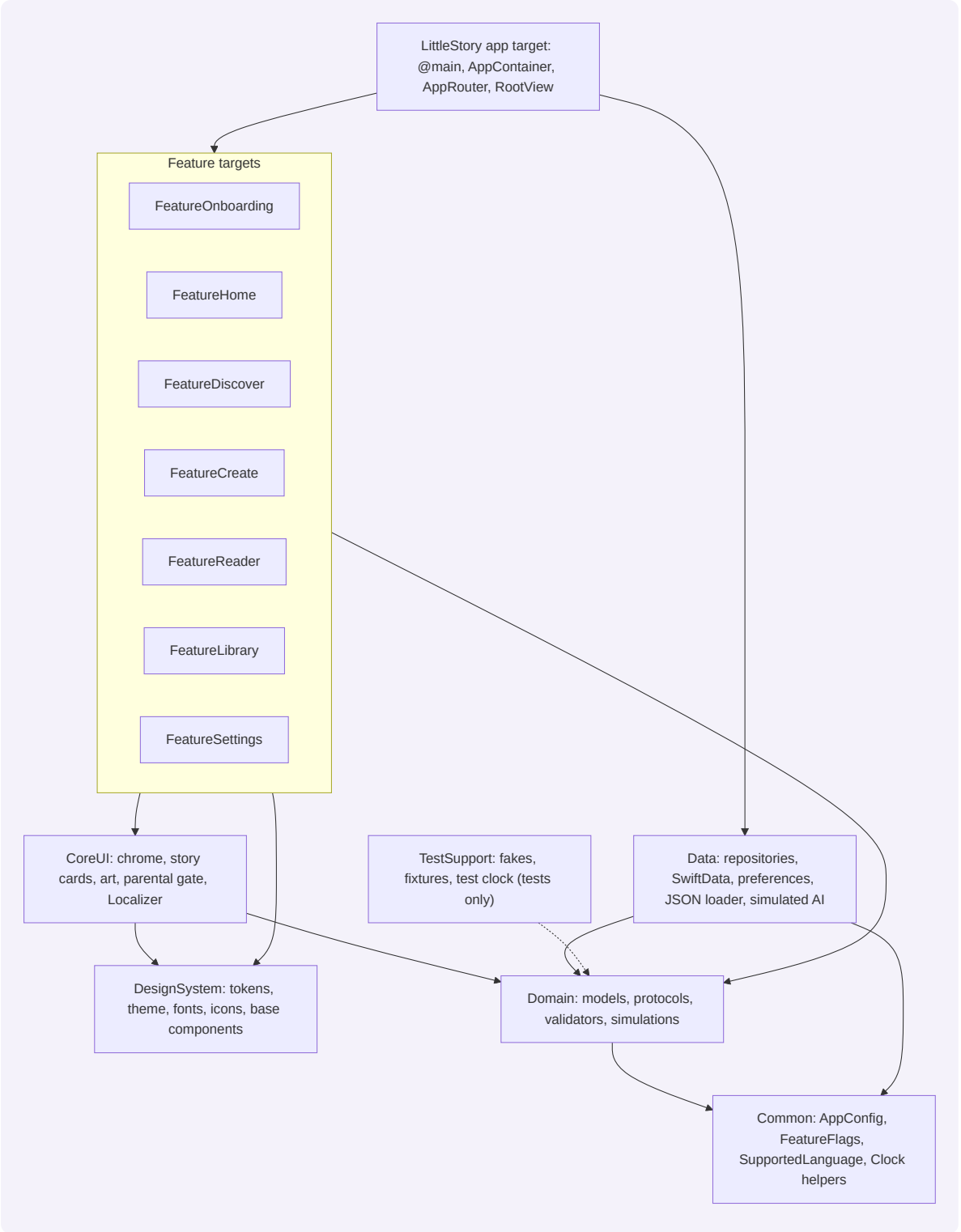


Layer	Knows about	Never knows about
Presentation	Domain protocols and models, design system, string catalogs	SwiftData, UserDefaults, JSON files, concrete repositories
Domain	Only Swift standard library and Foundation value types	SwiftUI, UIKit, SwiftData, localized text, bundles
Data	Domain protocols and models, storage frameworks	Views, view models, navigation

5.2 SOLID, applied

Principle	How it shows up
Single responsibility	A view only renders; a view model only maps intents to state; a repository only owns one kind of data; `PromptValidator` only validates
Open/closed	A new language, story source or AI provider is a new implementation or registry entry; screens do not change
Liskov substitution	`SimulatedStoryWriter`, a future `RemoteStoryWriter` and test fakes all satisfy the same `StoryWriter` contract and its tests
Interface segregation	Small protocols (`CatalogRepository`, `FavouritesRepository`…) instead of one big `StoryService`
Dependency inversion	View models depend on protocols in Domain; only `AppContainer` knows concrete types

5.3 Modules (targets in `LittleStoryKit`)



Android module	iOS target	Notes
`:app`	`LittleStory` app target	Composition root, root navigation, Info.plist, privacy manifest
`:core:common`	`Common`	Pure Swift
`:core:designsystem`	`DesignSystem`	Tokens generated from `design-tokens.json`

Android module	iOS target	Notes
`:core:ui`	`CoreUI`	Includes `Localizer` (replaces `UiText` + `AppLanguage`)
`:domain`	`Domain`	Pure Swift; port of the Kotlin models and rules
`:data`	`Data`	SwiftData replaces Room, `UserDefaults` replaces DataStore
`:feature:*`	`Feature*` (7 targets)	Each owns its views, view models, routes and strings
`build-logic` dependency rules	`Tools/check-dependencies.swift` + Package.swift	CI fails on a forbidden import

Dependency rules (enforced in CI):

1. Features never import other features or `Data`.
2. `Domain` imports only `Common` and Foundation (no SwiftUI, UIKit, SwiftData).
3. Only the app target imports `Data` and builds concrete objects.
4. `DesignSystem` imports nothing from the app.

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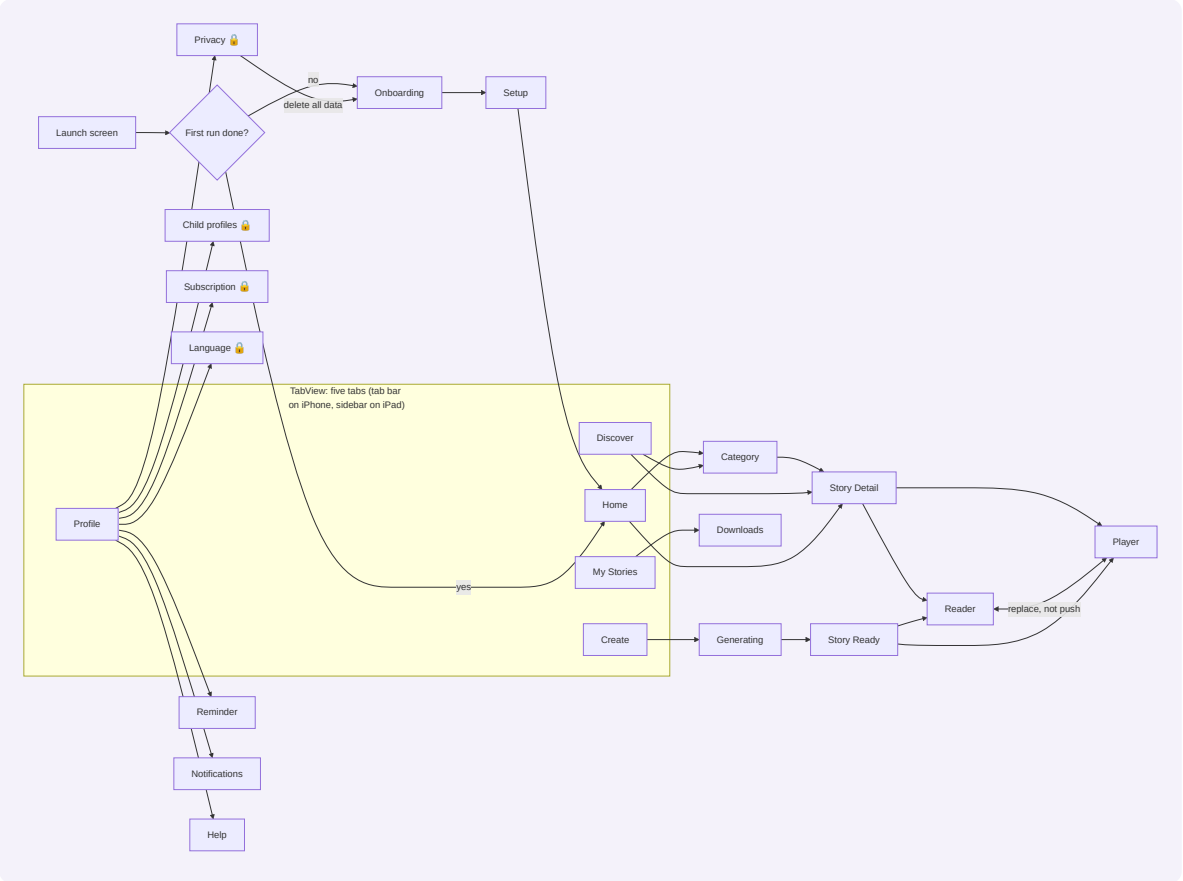
6. Screens and navigation

6.1 Screen list (same IDs as Android)

#	Screen	Feature	Parent only 🔒
01	Onboarding	Onboarding	
02	Setup (create / edit child)	Onboarding	
03	Home	Home	
04	Discover (search, filters)	Discover	
05	Category	Discover	
06	Story Detail	Discover	
07	Create	Create	
08	Generating	Create	
09	Story Ready	Create	
10	Reader	Reader	
11	Player	Reader	
12	My Stories (All, Created, Downloaded, Favourites)	Library	
13	Downloads	Library	

#	Screen	Feature	Parent only 🔒
14	Profile	Settings	
15	Subscription (demo)	Settings	🔒
16	Child profiles	Settings	🔒
17	Reminder	Settings	
18	Notifications	Settings	
19	Launch screen	App	
20	Language	Settings	🔒
21	Privacy and parent controls	Settings	🔒
22	Help	Settings	

6.2 Navigation map



- **Root:** `RootView` shows Onboarding or the `TabView` based on the first-run flag.
- **Tabs:** each tab has its own `NavigationPath` in a `TabRouter`, so switching tabs keeps each tab's history, like Android.
- **Typed routes:** each feature declares `enum XRoute: Hashable` and a `@ViewBuilder destination(for:)`; only the app target maps routes across features.
- **Reader ↔ Player** replace the last path element instead of pushing, so Back never loops.

- **Story Ready → Back** returns to Create's root (Create and Generating are removed from the path).
- **Deep links** (later) map a URL to a `[Route]` path in one place.

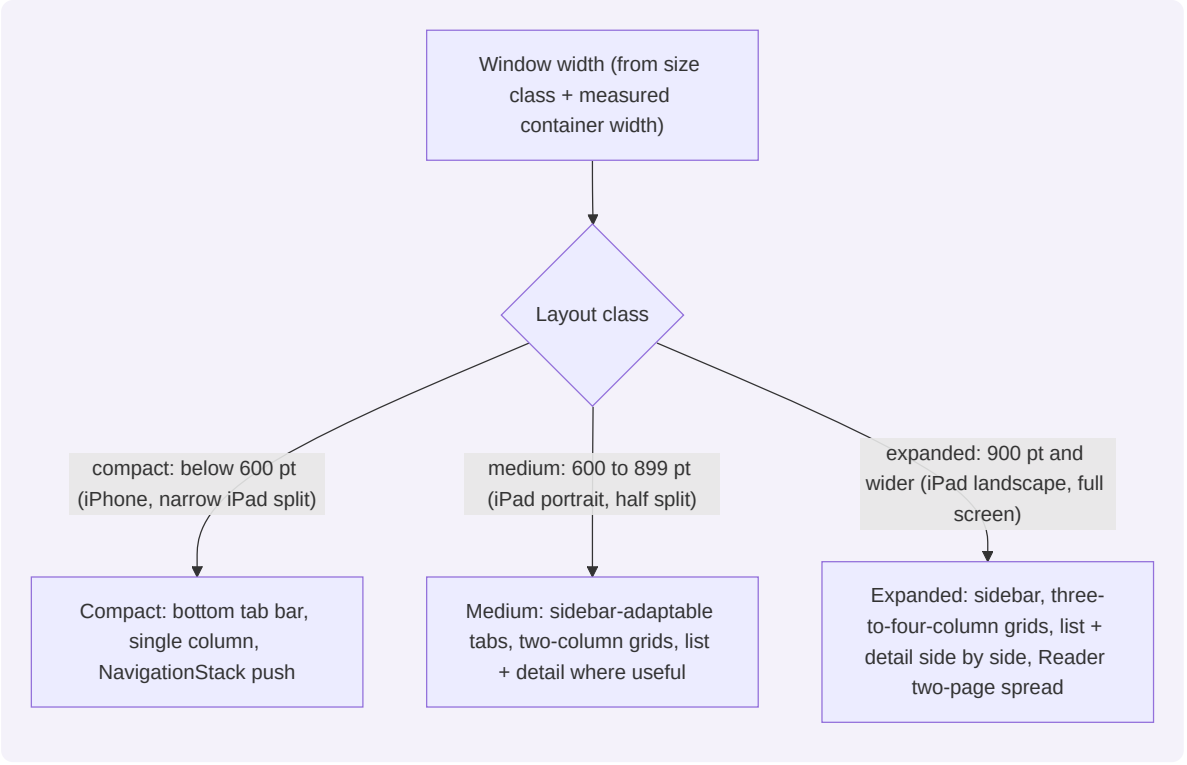
6A. Devices and adaptive layout (iPhone and iPad)

The app is **one universal binary**. Layouts are chosen by the space available, never by device model, so the same rules cover iPhone portrait, iPhone landscape, iPad split screen, Slide Over and freely resized iPad windows.

6A.1 Supported devices and modes

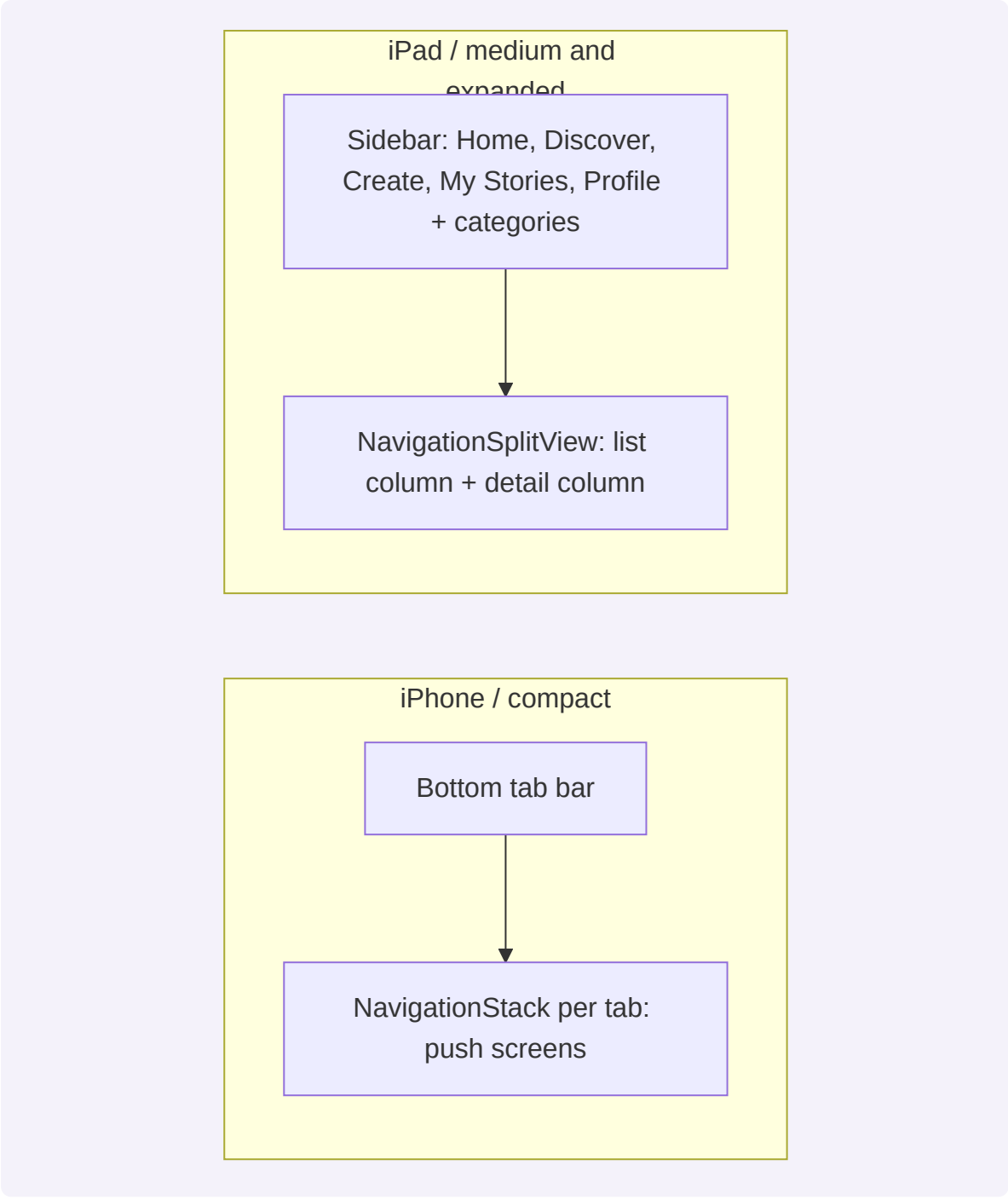
	iPhone	iPad
Oldest supported (iOS / iPadOS 26)	iPhone 11, 11 Pro, SE (2nd gen)	iPad (8th gen), iPad Air (3rd gen), iPad mini (5th gen), iPad Pro 11-inch (1st gen)
Orientation	portrait and landscape	all four orientations
Multitasking	—	Split View, Slide Over, Stage Manager, resizable windows (iPadOS 26 windowing)
Input	touch, VoiceOver, Switch Control	plus hardware keyboard shortcuts, trackpad pointer, Apple Pencil taps
Menu bar (iPadOS 26+)	—	`commands` for Create story, Search, Settings, Next / Previous page

6A.2 Layout classes



- Breakpoints are tokens (`LayoutTokens.mediumMinWidth = 600`, `expandedMinWidth = 900`) in `DesignSystem`, not literals in features.
- A screen reads one `LayoutClass` value from the environment; it is recomputed live when the window is resized or rotated.
- Content never stretches edge to edge on a wide window: readable text is capped at a token width (`readableMaxWidth`) and centred.
- Large Dynamic Type can push a screen to the compact layout even on iPad, so Kannada at AX sizes never truncates.

6A.3 App shell by layout class



`TabView` with `.tabViewStyle(.sidebarAdaptable)` gives the iPhone tab bar and the iPad sidebar from one definition; the person can still collapse the iPad sidebar to a top tab bar.

6A.4 Screen-by-screen layout

Screen	Compact (iPhone)	Medium / expanded (iPad)
Onboarding, Setup	full-screen pages	centred card, max width token, illustration beside the form when expanded

Screen	Compact (iPhone)	Medium / expanded (iPad)
Home	vertical sections, horizontal card rails	2–4 column card grids, Resume and Featured side by side
Discover	search + list, push to Category / Detail	split view: categories and results in the list column, Story Detail in the detail column
Story Detail	full screen	detail column, or a sheet from Home
Create	single form	form and live preview card side by side
Generating, Story Ready	full screen	centred, max width
Reader	one page, swipe	one page in portrait; **two-page spread** in expanded landscape; text width capped
Player	stacked cover + controls	cover and controls side by side; chapter list visible
My Stories, Downloads	segmented tabs + list	grid of cards; filters in the sidebar
Profile and settings	grouped list, push	split view: settings list + selected setting
Parental gate	sheet	form sheet sized for the keypad, centred

6A.5 Design deliverables this needs

The current design reference (`docs/design/screens/*.jpg`) is phone-only. To keep tablet layouts on-brand:

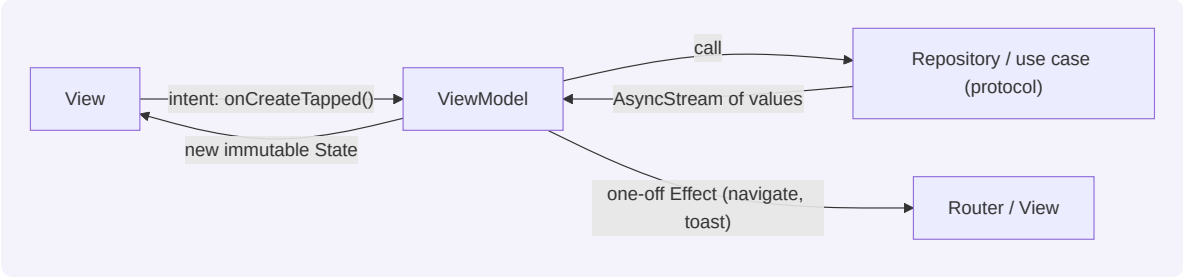
1. Designer (or owner) adds iPad frames for Home, Discover + Detail, Reader spread, Player and Settings split view in the Figma file, using the same tokens.
2. Until those exist, the rules in §6A.4 are the specification, and the iPad snapshot images are reviewed by the owner as the design check.
3. Token additions: layout breakpoints, grid column counts per class, readable max width, sidebar width; all in `design-tokens.json` so Android tablets can use the same values later.

6A.6 Liquid Glass (iOS 26 design)

- System bars, tab bar, sidebar, sheets and toolbars use the system Liquid Glass material automatically; we do not re-skin them.
- Brand colour and illustrations stay in content; glass is used only for floating controls (Player controls, Reader toolbar) through the design system, never directly in features.
- Respect Reduce Transparency and Increase Contrast settings: the design system swaps glass for solid token colours.

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7. Presentation pattern (unidirectional data flow)



1. Each screen has **one immutable `State` struct** owned by an `@Observable` `@MainActor` view model.
2. The view calls **one method per user action** (`onNext()`, `onToggleFavourite()`), matching the Android pattern.
3. View models **observe repositories as `AsyncSequence`s** in a `.task` tied to the view's lifetime, so work stops when the screen goes away (the iOS equivalent of `collectAsStateWithLifecycle`).
4. **One-off effects** (navigate after generation, data deleted) are delivered through an `AsyncStream<Effect>` or a callback closure to the router; they are never stored in `State`.
5. **Views are split** into `XScreen` (takes `State` + `Actions`, no view model, used by previews and snapshots) and `XRoute` (creates the view model and wires navigation).
6. View models **never hold display strings**; they hold `LocalizedStringResource` values or enums, resolved by the view.

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8. Data and persistence

Store	Technology	Contents	Child data?
Mock content	JSON in the `Data` target's resources, decoded with `Codable`	`stories_{en,kn}`, `categories_{en,kn}`, `generated_{en,kn}`, `notifications_{en,kn}`, `plans` (identical files to Android)	No
`LittleStory.store`	SwiftData, versioned schema	child profiles, saved stories, reading progress, downloads, generated stories, notification read state	First name, age, avatar, interests only

Store	Technology	Contents	Child data?
Preferences	`UserDefaults` (app suite)	first-run flag, content language, active child id, text scale, playback speed, reminder, notification and download settings	No
UI language	iOS per-app language (`AppleLanguages`) or in-app override (§9)	language tag	No

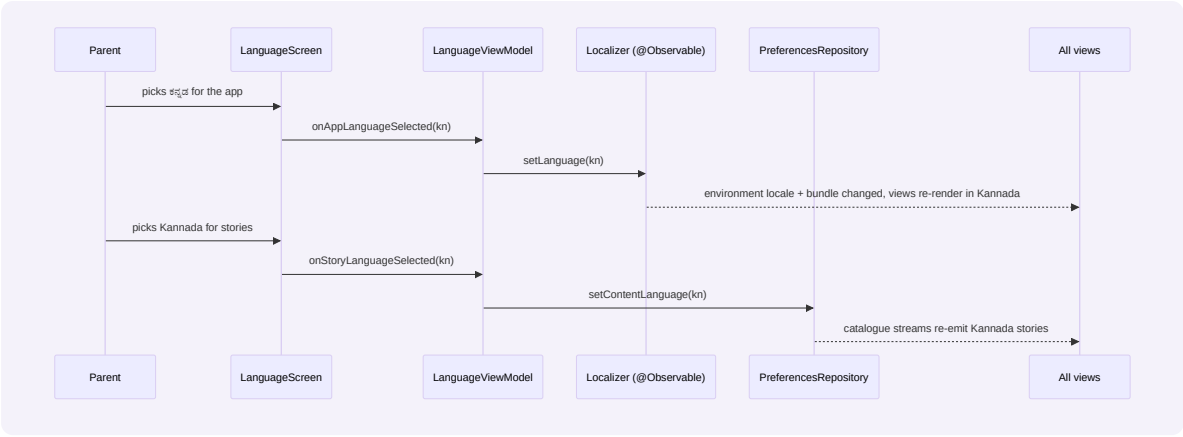
- **Single source of truth:** the store emits, every observing screen updates; view models never keep a second copy of persisted state.
- **English is the base** for all content; a translation file overrides text field by field, so a missing Kannada field shows English instead of failing.
- **Delete all data** wipes the SwiftData store and preferences and returns to Onboarding.
- **File protection:** the store uses `NSFileProtectionComplete` (`.completeUntilFirstUserAuthentication` for background-safe access) and is excluded from iCloud backup until accounts and consent exist.

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9. Localization

Topic	Rule
Languages	`SupportedLanguage` registry in `Common` : `en` (default, fallback), `kn` . Native names shown in pickers: English, ಕನ್ನಡ
UI strings	One `Localizable.xcstrings` per target that owns UI; no literal user-facing text in Swift
View model text	`LocalizedStringResource` or enums mapped to resources (the iOS `UiText`)
Plurals and arguments	String Catalog plural variants and positional `%1$@` / `%2$lld` ; never concatenate sentences
Two languages, two settings	**UI language** (menus, buttons) and **story language** (content) are separate, like Android
Switching UI language	Primary: an in-app `Localizer` override that reloads strings at once without restarting. Also supported: the system per-app language page in iOS Settings, opened from the Language screen
Content	Repositories take a `languageTag` and resolve with English fallback
Layout	Leading/trailing only, no fixed text heights, multi-line everywhere, tested at the largest accessibility size with Kannada
Numbers and dates	`FormatStyle` with the active locale

Topic	Rule
Fonts	Bundled Noto Sans Kannada with taller line height for Kannada
Gates	CI fails if any key lacks a Kannada value or is marked "needs review" without a workbook entry; new Kannada text is a draft until a native speaker approves it



> **Spike LS-iOS-S1:** prove the in-app override in a throwaway branch (String Catalog lookup with an explicit locale and bundle across all targets, plus `\.locale` for formatting). If any piece cannot follow the override, fall back to the system per-app language page only.

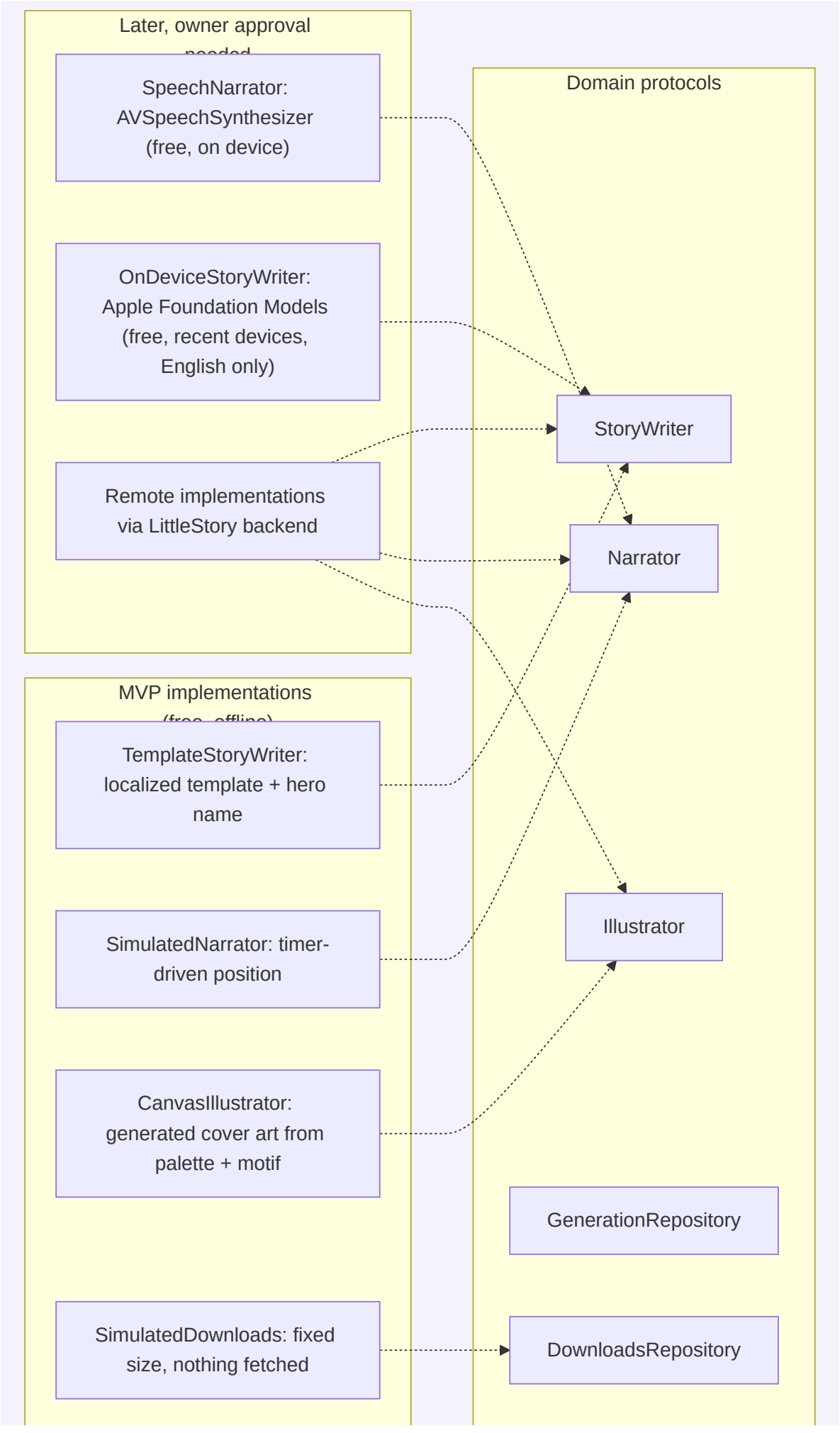
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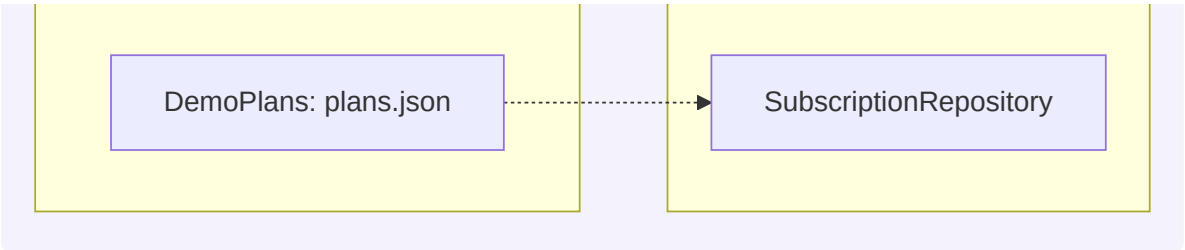
10. Keeping iOS and Android in step (shared contracts)

Contract	Source of truth	How iOS uses it
Mock content JSON	Android `data/src/main/assets/mock/`	Copied into `Data/Resources/Mock/` by `Tools/sync-android-content.sh`; same decoder tests on both sides
Design tokens	`docs/design/design-tokens.json`	`Tools/gen-tokens` generates `DesignSystem/Tokens.generated.swift`
Translations	Android `strings.xml` (EN, KN) and the workbook Translations sheet	`Tools/import-android-strings` converts to `.xcstrings`, keeping the same keys
Screens and states	`docs/design/screen-inventory.md`, `screen-state-matrix.md`	Same screen IDs, same states, same acceptance criteria
Test cases	Workbook test cases (TC-###)	iOS tests reference the same IDs
Backlog	Jira workbook	iOS tickets use the `LS-iOS-###` prefix

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11. Simulated (mocked) AI features


Later, owner approval needed
SpeechNarrator: AVSpeechSynthesizer (free, on device)
OnDeviceStoryWriter: Apple Foundation Models (free, recent devices, English only)
Remote implementations via LittleStory backend
MVP implementations (free, offline)
TemplateStoryWriter: localized template + hero name
SimulatedNarrator: timer-driven position
CanvasIllustrator: generated cover art from palette + motif
SimulatedDownloads: fixed size, nothing fetched
Domain protocols
StoryWriter
Narrator
Illustrator
GenerationRepository
DownloadsRepository



Feature	MVP simulation	UI label	Swap point
Story generation	Progress +4 % every 120 ms in three stages, then the localized template filled with the hero's first name and saved	"Demo story" badge	`StoryWriter`
Narration	Position advanced on a 500 ms tick; chapters = pages; speed, skip 15 s, sleep timer	"Narration preview, no sound yet"	`Narrator`
Pictures	Generated cover art from the story's palette and motif (SwiftUI `Canvas`)	none (art is by design)	`Illustrator`
Downloads	A saved record with a fixed size	"Simulated download"	`DownloadsRepository`
Subscription	Illustrative plans, no purchase	"Demo plans"	`SubscriptionRepository`
Notifications	Mock inbox per language, nothing scheduled	—	`NotificationsRepository`

- All simulations take an injected `Clock`, so tests run them instantly.
- Selection happens in one place: `AppContainer` reads `FeatureFlags` and picks the implementation.
- The prompt the parent types is never stored, matching Android.

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12. Cross-cutting concerns

Concern	Approach
Configuration	`AppConfig` (limits, options, timings) and `FeatureFlags` in `Common`; build settings in `.xcconfig` per configuration (Debug, Release); no secrets anywhere
Design system	Tokens for colour, type, spacing, radius, elevation, motion; components (`LSButton`, `LSChip`, `LSTextField`, banners, toasts); one light theme with night surfaces for bedtime screens
Accessibility	44 pt minimum targets (Apple HIG), VoiceOver labels and hints from string catalogs, `accessibilityValue` for progress, Dynamic Type up to AX5, Reduce Motion respected, contrast AA

Concern	Approach
Errors	No error reaches the user as a crash or raw text: missing content becomes a Not Found state, bad JSON falls back to English, an unknown generation request becomes a paused state
Logging	`os.Logger` per subsystem; no child names or prompts in logs
Security	No network entitlement needed beyond default, ATS left on, no third-party SDKs, data protection on files, no secrets in the repo (public)
Privacy	`PrivacyInfo.xcprivacy` declares no tracking, no collected data, and the required-reason API for `UserDefaults` (CA92.1)
Kids category readiness	Parental gate before links out, purchases and settings (App Store Review Guidelines 1.3 and 5.1.4); no ads, no analytics
Performance	Lazy stacks for lists, JSON parsed once per language and cached in an actor, images drawn not decoded, launch < 1 s on the oldest supported devices (iPhone 11, iPad 8th gen)

Cost and tooling notes (no paid services)

Need	Free path	Paid only if...
Build and run in Simulator	Xcode on any Mac	—
Run on your own iPhone	Free Apple ID "personal team" signing; the app expires after 7 days and must be re-installed	—
CI	GitHub Actions macOS runners are free for public repos	the repo is private (minutes are charged at a 10× rate after the free quota)
TestFlight, App Store	—	**Apple Developer Program, US$99 / year**; owner decision at release time
AI and voices	Mocks; optional on-device AVSpeechSynthesizer	real AI or premium voices

> **Important:** iOS apps can only be built with Xcode on macOS. The cloud agent workspace is Linux, so iOS build and test evidence must come from GitHub Actions macOS runners or the owner's Mac.

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13. Repository, branching and CI

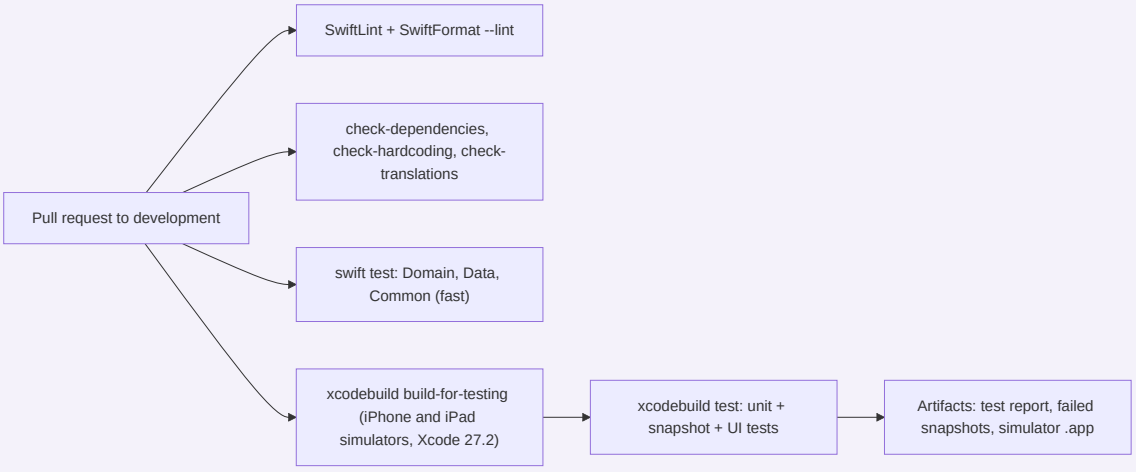
13.1 Repository layout

```
LittleStory-iOS/
├── LittleStory.xcodeproj          # thin app project
├── App/                           # @main, AppContainer, AppRouter, RootView,
Info.plist, PrivacyInfo.xcprivacy, Assets
├── Config/                        # Debug.xcconfig, Release.xcconfig,
Shared.xcconfig
├── Packages/LittleStoryKit/       # Package.swift + Sources/<Target>/ +
Tests/<Target>Tests/
├── Tools/                         # check-dependencies, check-hardcoding,
check-translations, gen-tokens, sync scripts
├── docs/                          # HLD, LLD, ADRs, localization, testing
├── .github/workflows/ci.yml
├── .swiftlint.yml, .swiftformat, .xcode-version, .gitattributes, .gitignore
└── CLAUDE.md, .github/copilot-instructions.md
```

13.2 Branching (same model as Android)

`development` (default, PR target) → `staging` → `main`; `feature/LS-iOS-###-slug`, `fix/…`, `docs/…`, `chore/…`; only the owner merges promotions and releases.

13.3 CI pipeline

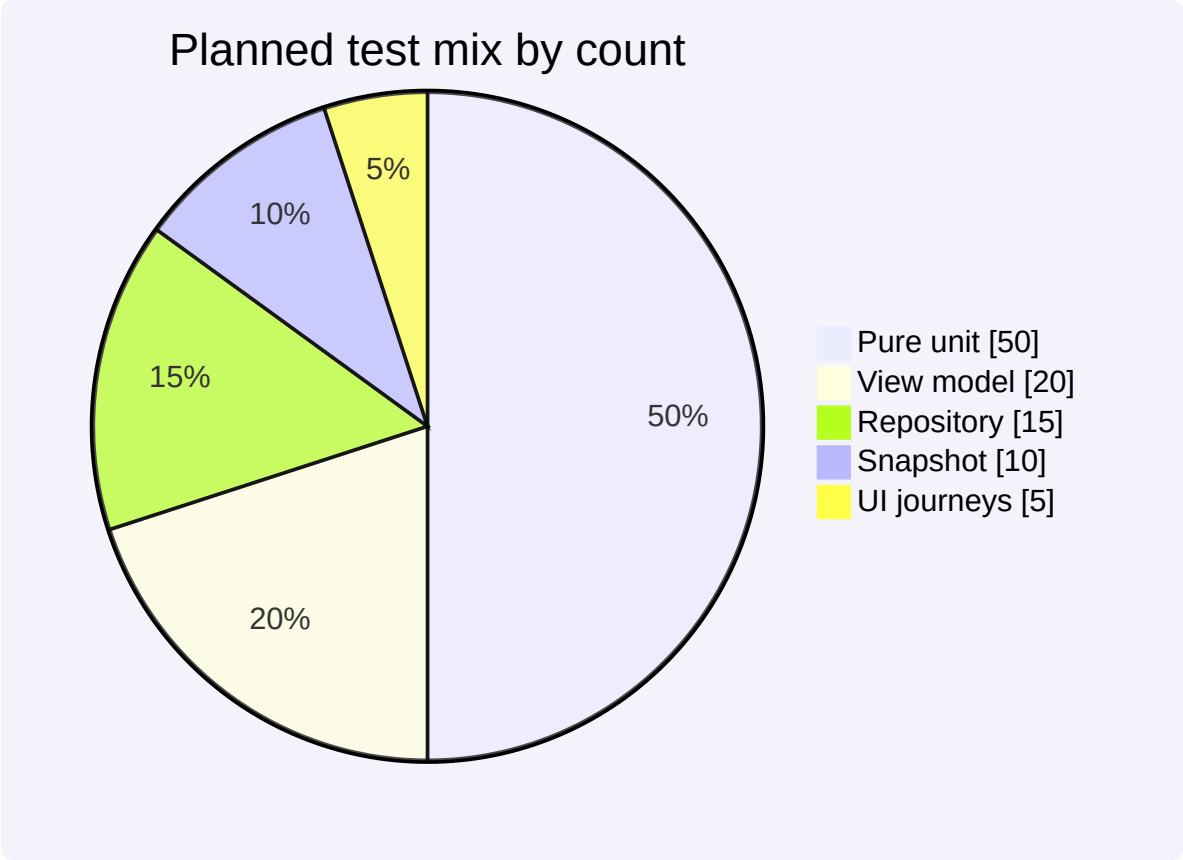


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14. Testing strategy

Level	Tool	What	Target share
Pure unit	Swift Testing	validators, simulations, config, content resolver	50 %

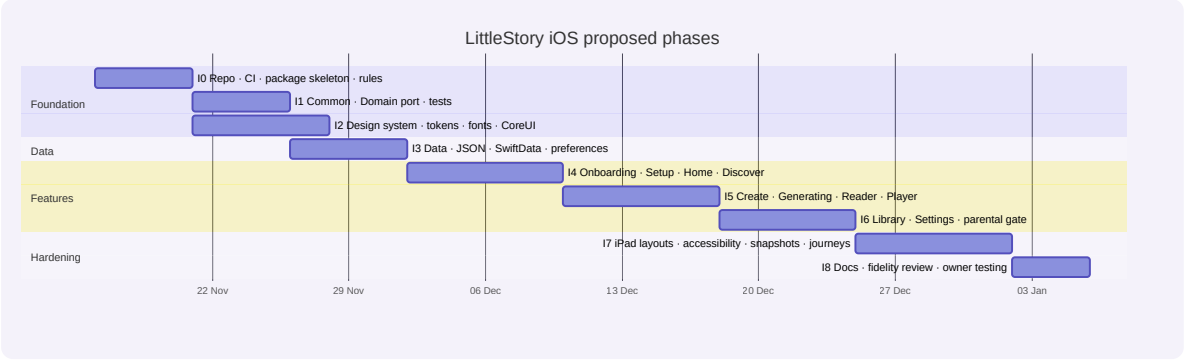
Level	Tool	What	Target share
Repository	Swift Testing + in-memory SwiftData	catalogue merge, persistence, favourites ordering	15 %
View model	Swift Testing + fakes + test clock	every screen's state transitions and effects	20 %
Snapshot	swift-snapshot-testing	each screen × iPhone + iPad (portrait, landscape) × EN / KN, plus KN at AX5	10 %
UI journey	XCUITest	first run, create story, read, listen, switch language, delete data	5 %



- Coverage gate: ≥ 80 % lines on Domain and view models (`xccov` report in CI).
- UI tests find elements by accessibility identifier, never by English text.
- Snapshot baselines are approved by the owner, never auto-approved.

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15. Delivery plan



Phase	Exit criteria
I0	Empty app builds in CI; dependency, lint and translation gates run
I1	Domain port passes tests mirroring Android `:domain` tests
I2	Token-driven components with snapshots in EN / KN
I3	Repositories pass the same scenarios as Android `:data` tests
I4–I6	Each screen matches its design reference and state matrix on iPhone, and has a working compact and regular layout, with tests
I7	iPad split view, sidebar, Reader spread and resized windows reviewed; AX5 Dynamic Type and VoiceOver pass on every screen; journeys green on iPhone and iPad
I8	Owner tests on iPhone and iPad (Simulator or own devices); owner decides on Developer Program

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16. Risks and open questions

#	Risk or question	Mitigation / owner action
R1	In-app language switch without restart is not a standard iOS API	Spike S1 (§9); fallback to the system per-app language page
R2	Kannada text overflows at large Dynamic Type	Multi-line layouts, snapshot at AX5 in Kannada as a gate
R3	SwiftData maturity (migrations, concurrency)	Hidden behind repository protocols; GRDB fallback (LLD §4.4)
R4	Android and iOS drift	Shared contracts and sync tools (§10); same test IDs
R5	No Mac in the cloud workspace	All evidence from CI macOS runners or the owner's Mac
R6	Store release needs a paid Apple account	Owner decision at release (Level F); not needed for development

#	Risk or question	Mitigation / owner action
R7	No iPad designs exist yet	§6A rules are the spec; owner reviews iPad snapshots; iPad frames added to Figma when possible
R8	GitHub's `macos-26` runner defaults to Xcode 26.6	Workflow selects Xcode 27.2; if the image lacks it, use the `xcode-27` runner label until it is added
Q1	~~Confirm native SwiftUI (D-iOS-1)~~	Decided 11 Nov 2026: native Swift + SwiftUI
Q2	Repository name and visibility (public keeps CI free)	Owner; proposed `LittleStory-iOS`, public
Q3	Minimum iOS 26 acceptable?	Owner; iOS 26 and 27 run on the same devices, so iOS 26 reaches everyone who can run the latest OS while keeping the newest APIs
Q4	~~iPad scope~~	Decided 11 Nov 2026: iPhone and iPad, adaptive layouts from the start (§6A)
Q5	Bundle identifier, e.g. `com.littlestory.app`	Owner